

Executive Summary for the Manager at the Development Office

Main Question: What are the main drivers of donating in 2023, how accurately can the probability of donating be estimated, and is 3rd party information worth it?

For the year 2024, I recommend targeting alumni with the following characteristics:

- Donated for more than 3 years
- Bequest likelihood greater than 700
- Donated in the last decade
- Annuity likelihood greater than 700
- Donated some amount in 2023

Alumni characteristics that contribute most to the model used can be found in **Table A** below.

Table A:

Characteristic	Contribution to Model (Importance)
Number of Years Donated	26.4%
Bequest Likelihood	11.5%
Total Cash Donated	8.7%
Annuity Likelihood	6.2%
Amount Donated in 2023	4.7%
Total Contribution	57.5%

The model used to understand main drivers and predict the likelihood an alumnus donates has a 93.2% accuracy. Bequest Likelihood and Annuity Likelihood are alumni metrics from a 3rd party institution that contribute to the model's performance by 17.7% making it an integral aspect of the model.

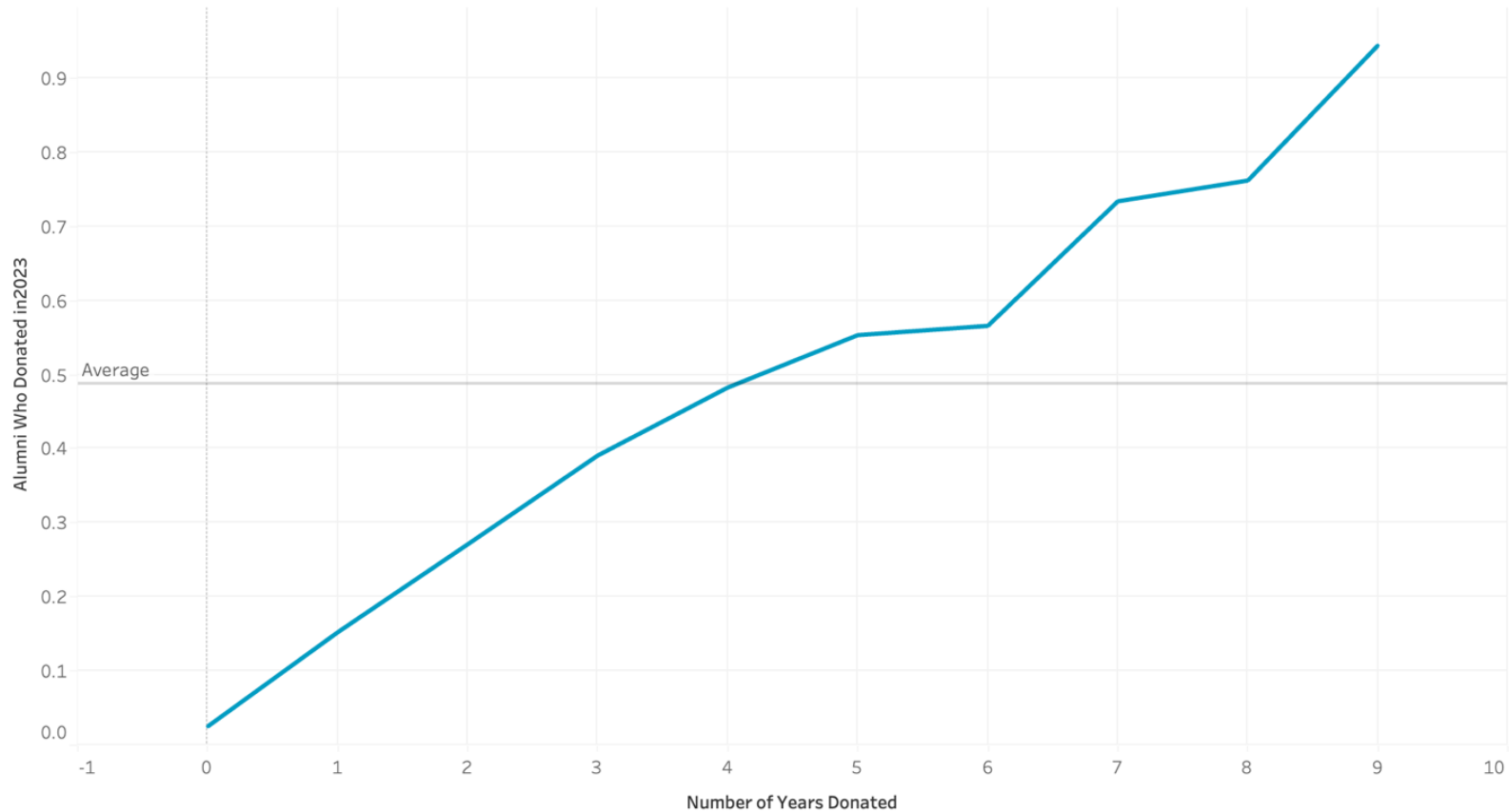
This data comes from The University of Tennessee and the following metrics comes from a 3rd party institution: Bequest Likelihood, Annuity Likelihood, CRT Likelihood, EstTotalAssets, EstNetWorth, EstIncome, EstGiftCapacity. This data is from the year 2023 and is comprised of 30,000 individuals (rows) and 53 characteristics (columns). The characteristics are composed of basic demographic information, donation history, degrees earned and more.

Since this data is from 2023, the model created is based on donor behavior that takes place in the year of 2023. However, donor behavior may change in future years and this data does not account for that. Additional data that would improve results is including a characteristic that demonstrates alumni satisfaction based upon a survey.

Discussion of Findings

Analyzing Main Drivers – Number of Years Donated

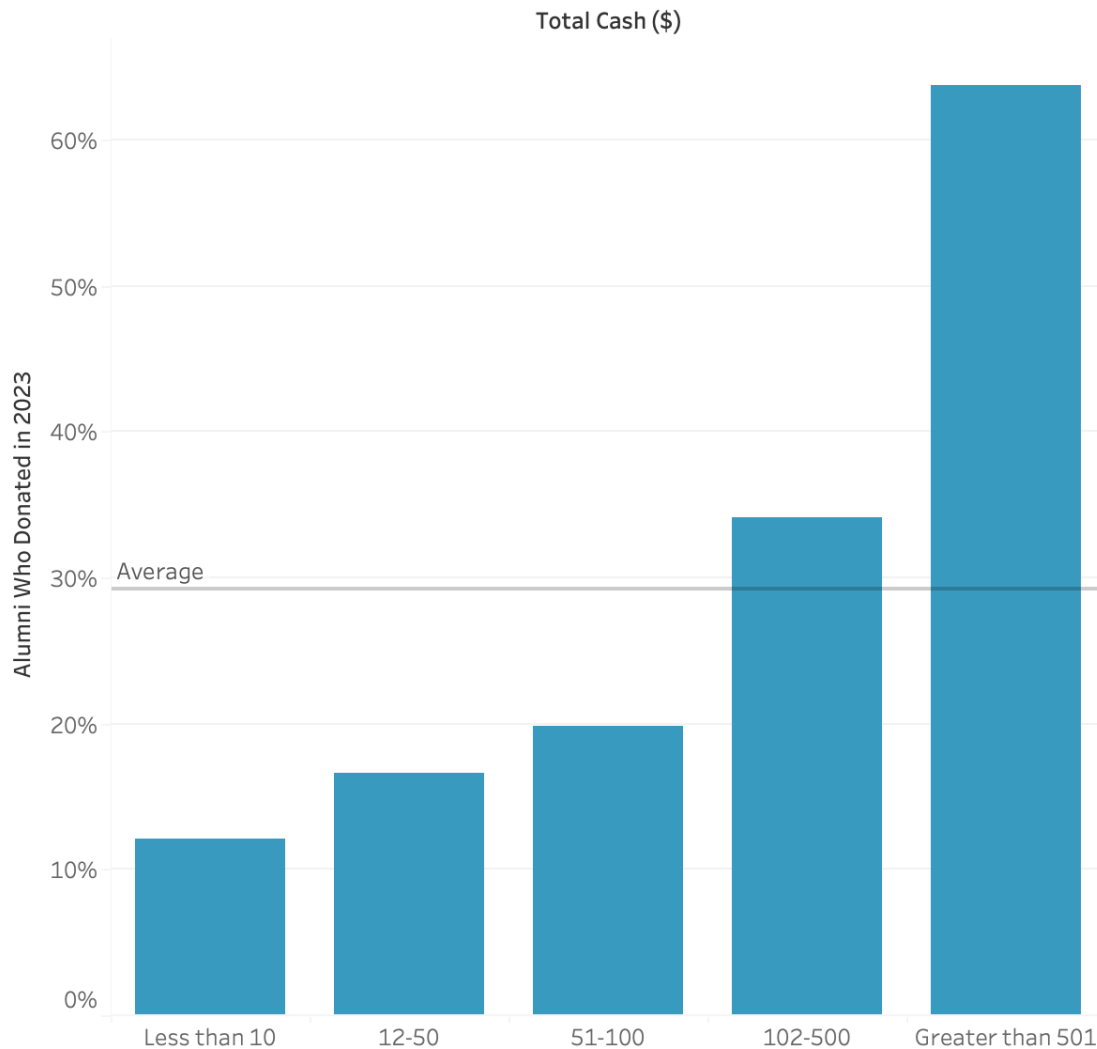
Alumni Donation by Number of Years Donated



The number of years an alumnus has donated has a positive relationship with the alumnus donating. Alumni who have donated for four or more years continue to donate higher than the average alumnus

Analyzing Main Drivers – Total Cash Donated

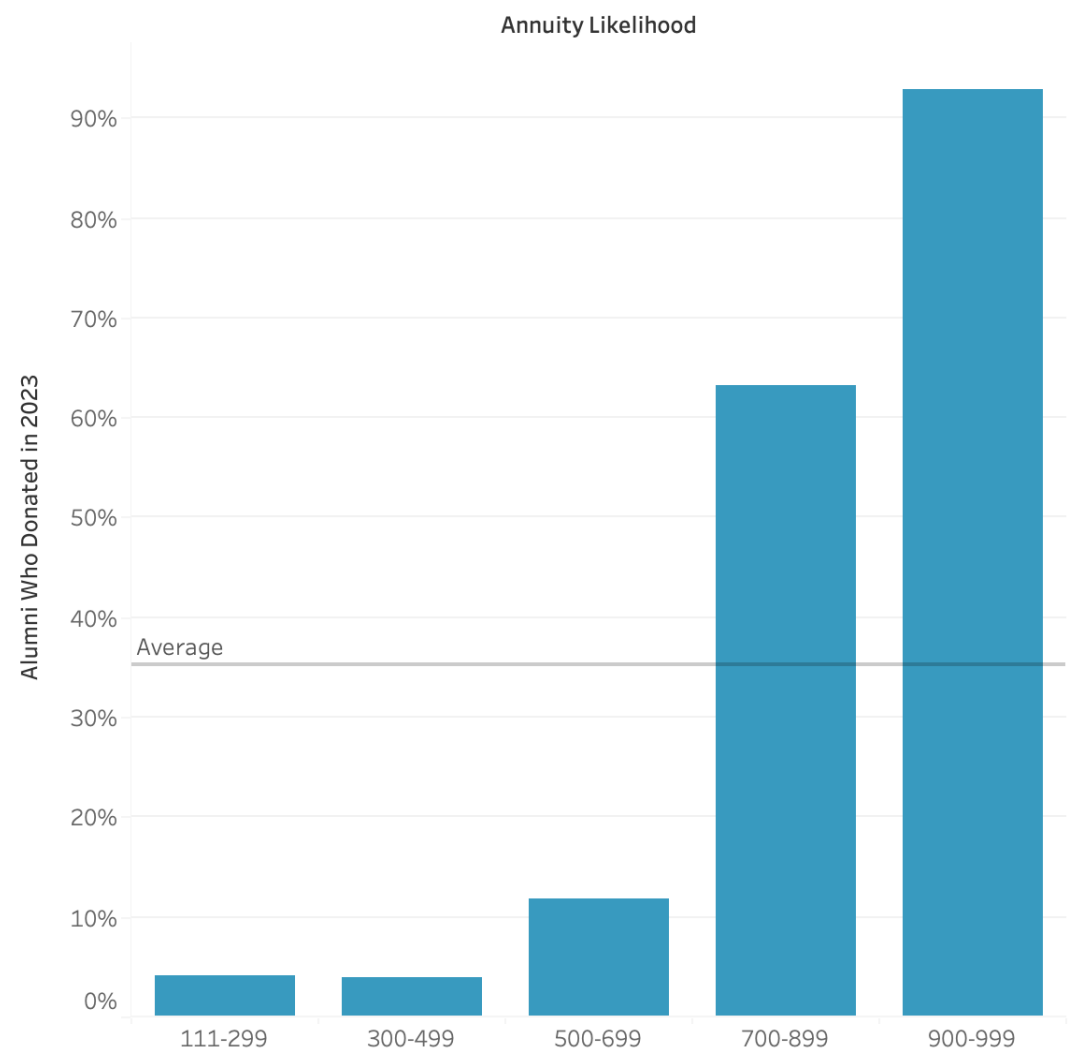
Alumni Donation by Total Cash Donated in the Last Decade



If an alumnus has donated greater than 102 in the last decade, they are more likely to donate in 2023. This likelihood of an alumnus donating in 2023 double if the alumnus' total cash donation in the last decade is greater than 501

Analyzing Main Drivers – Annuity Likelihood

Alumni Donation by Annuity Likelihood



Alumni with an annuity likelihood metric greater than 700 are more likely to donate in 2023 than those with an annuity likelihood below 700.

Analyzing Main Drivers – Bequest Likelihood

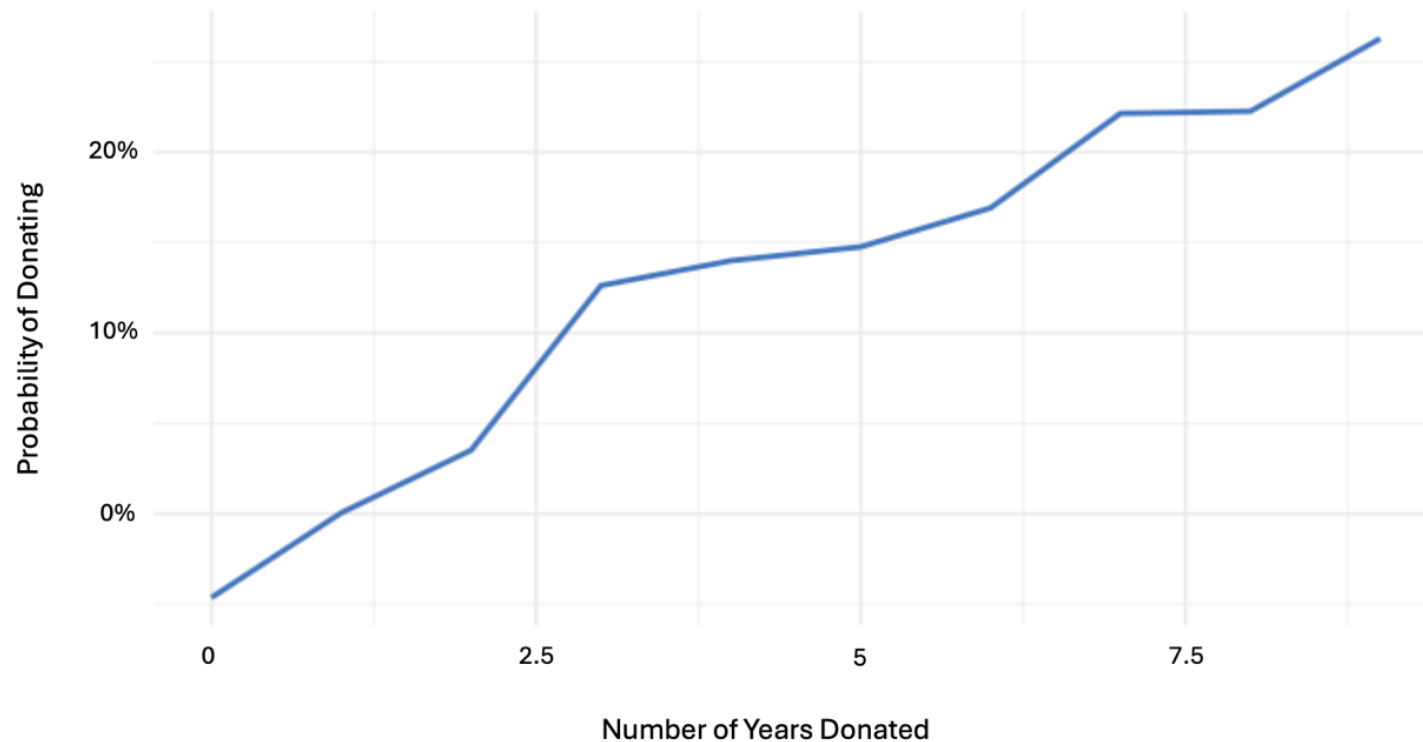
Alumni Donation by Bequest Likelihood

Bequest Likelihood	Percentage of Alumni that Donated
118-199	8%
200-299	5%
300-399	5%
400-499	5%
500-599	7%
600-699	12%
700-799	31%
800-899	75%
900-999	90%

Alumni with a bequest likelihood metric greater than 800 are more likely to donate in 2023 than those with a bequest likelihood below 800. Alumni with this characteristic have a greater than 75% likelihood of donating in 2023 compared to those with a bequest likelihood of under 800 who would have a donation likelihood equal to 31% and below.

Partial Dependence Plot – Number of Years an Alumnus Has Donated

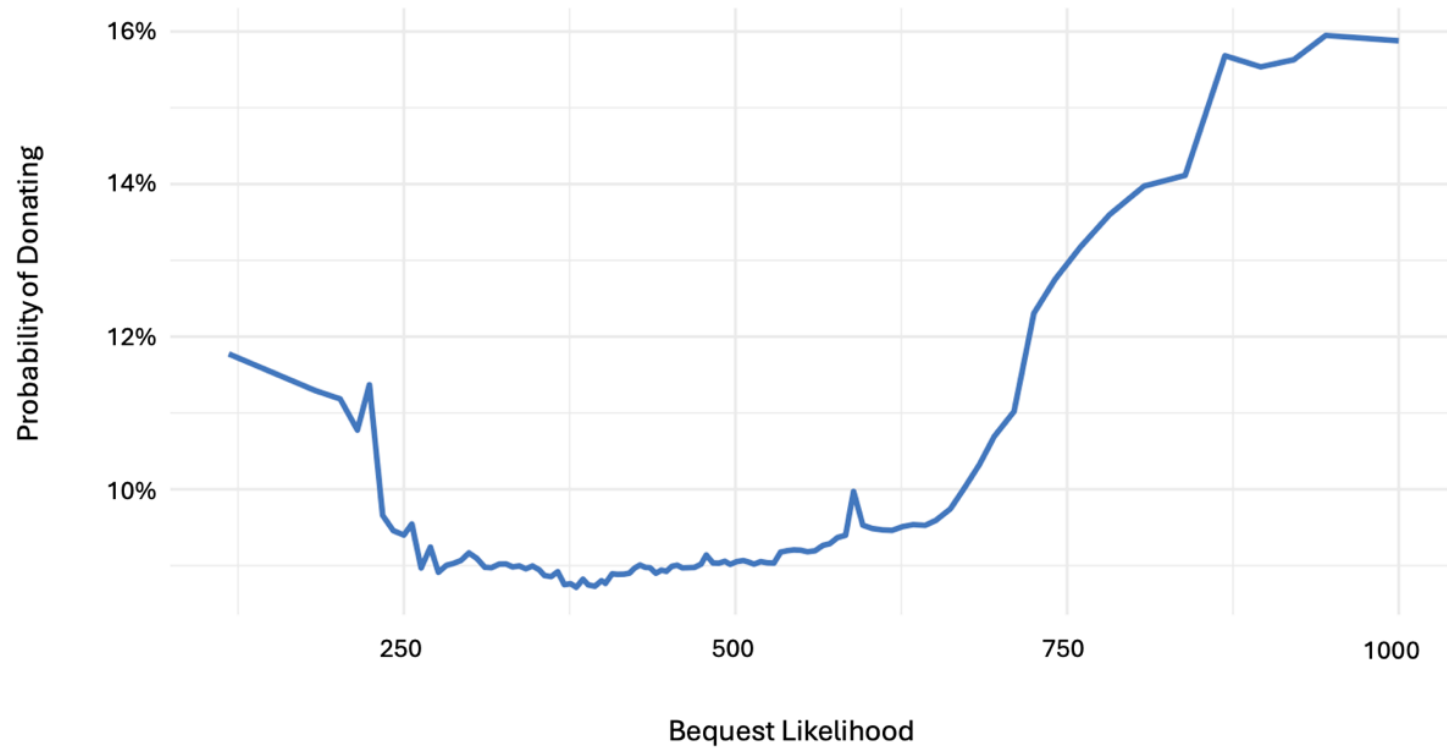
Accumulated Partial Dependence Plot for Number of Years Donated



Here we can see the impact of the number of years an alumnus has actively donated has on the probability of them donating again in 2023. The longer an alumnus has actively donated, the more likely they were to have donated again in 2023. We can see the magnitude of this characteristic as the probability of donating nearly double when the number of years donated jumps from 2.5 years (8%) to 5 years (15%).

Partial Dependence Plot – Bequest Likelihood

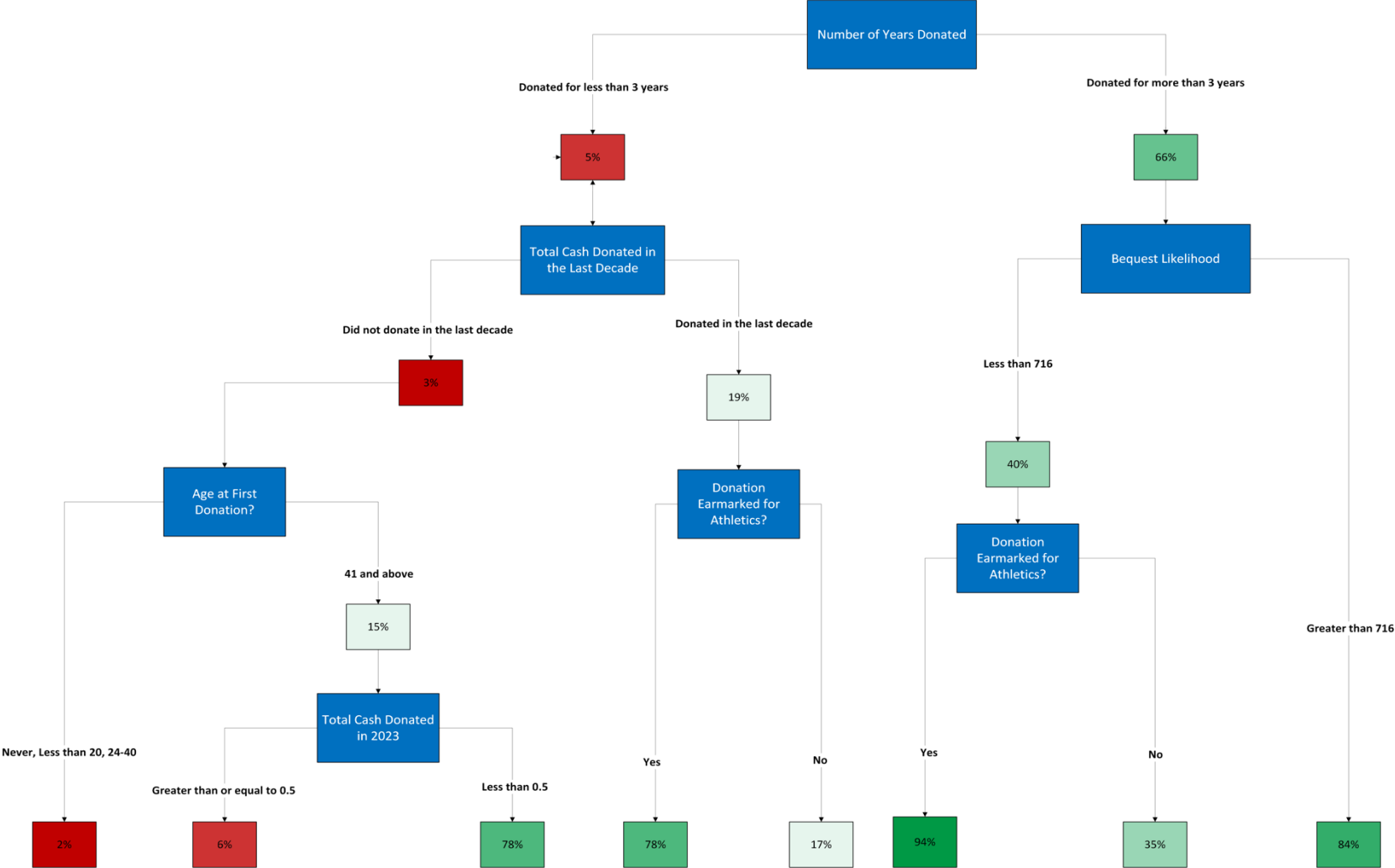
Accumulated Partial Dependence Plot for Bequest Likelihood Metric
Provided by Third Party Institution



A higher bequest likelihood leads to a higher probability of the alumnus donating. However, the probability levels out and becomes stagnant at a bequest likelihood of 825. The probability of donating over doubles when the bequest likelihood jumps from 500 to 750. We can see the magnitude of bequest likelihood using this plot since it is isolated.

Analyzing Combinations – Decision Tree

Measuring the Likelihood of Alumni Donation



Discussion: Measuring the Likelihood of Alumni Donation Decision Tree

The decision tree shows how various combinations of alumni characteristics result in a higher or lower donation predicts. Some of the characteristics that were included in the decision tree are the following:

- Number of years the alumni has donated
- Total cash donated in the last decade
- Bequest Likelihood (from a 3rd party institution)
- Age at first donation
- Total cash donated in 2023
- Whether the donation will be earmarked for athletics

We can see if an alumnus has actively donated for more than three years, then their chances for donating again are well above the 12.7% average donation rate. Combining this with characteristics like Bequest likelihood and whether the donation has been earmarked for athletics will increase or decrease the predicted chance of donation.

If an alumnus has not donated or has been actively donated for less than three years, then their predicted chance of donation is generally below average. Combining this with certain characteristics will increase or decrease this prediction. Particularly, if an alumnus had not donated in the past decade but are above the age of 41, their predicted chance of donating is dependent on whether they had already donated this year. If they have, their chance of donating is 6%. However, if they did not yet donate then their chance of donating is 78%.

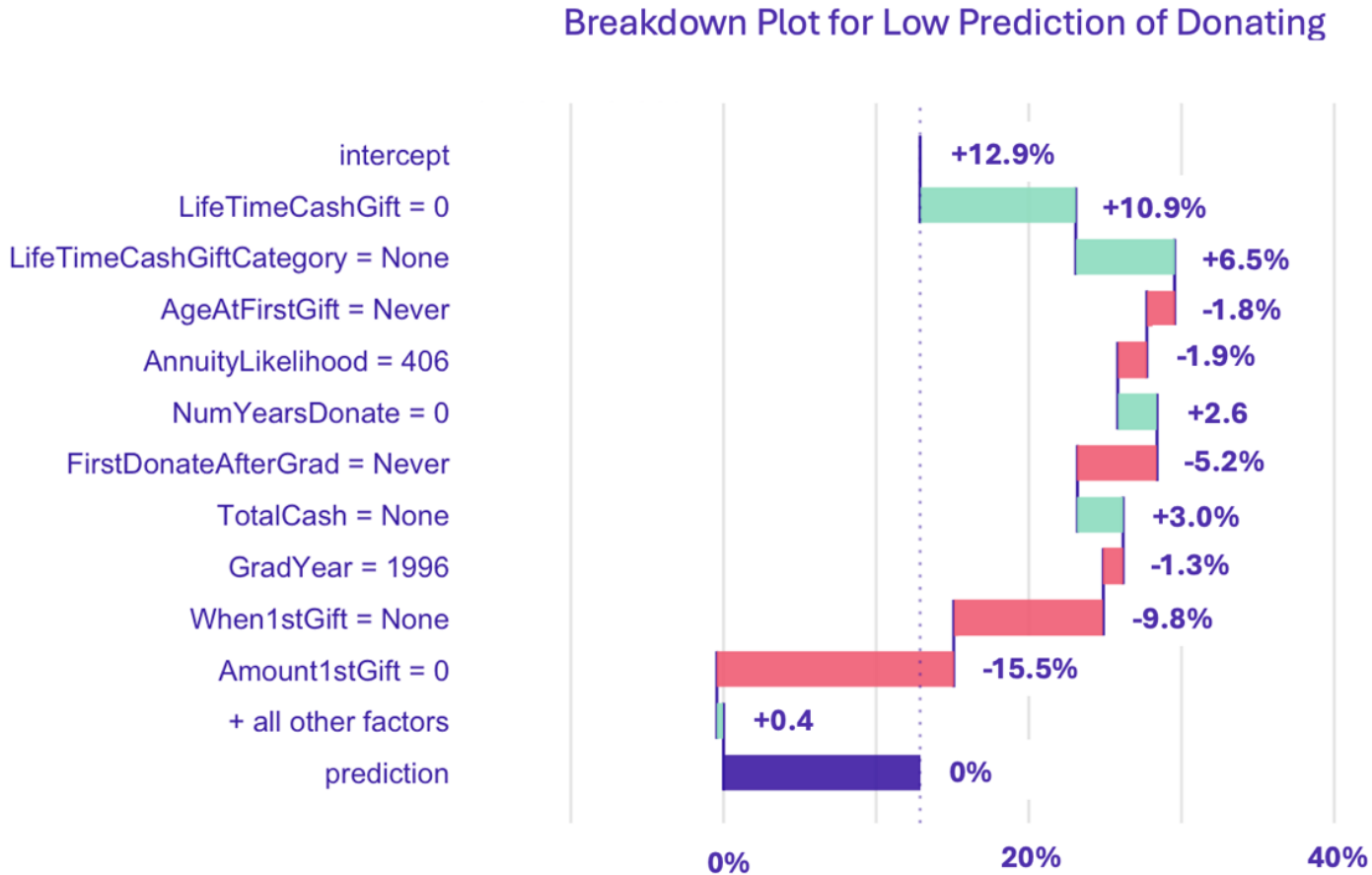
Combination with the highest predicted donation – 94% :

- Donated for more than 3 years
- Bequest Likelihood is less than 716
- Donation is earmarked for athletics

Combination with the lowest predicted donation – 2%:

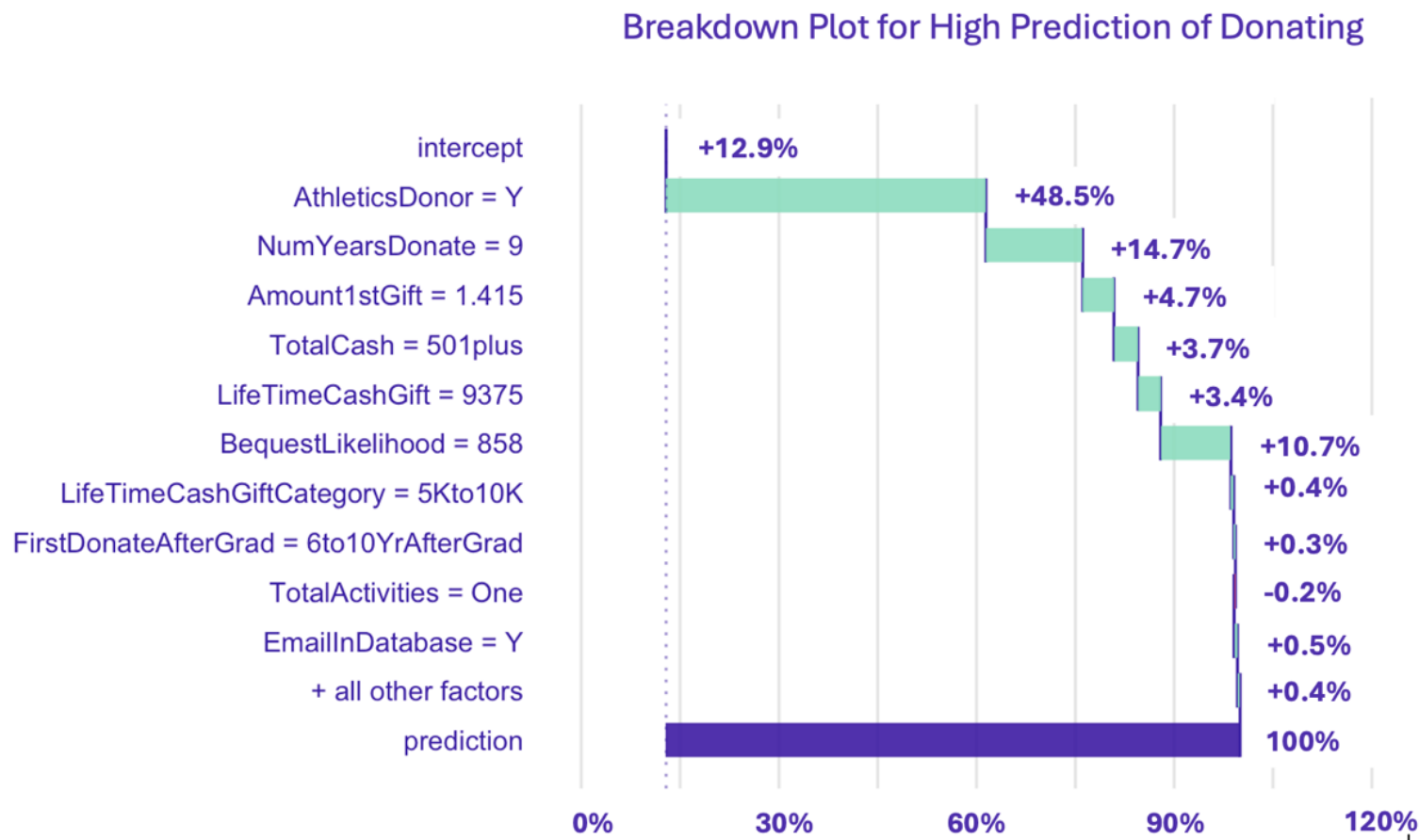
- Donated for less than 3 years
- Did not donate in the last decade
- Never donated or age at first donation is less than 20 or 24-40

Breakdown Plot – Low Likelihood of Donating Prediction



The characteristics displayed in the breakdown plot combine to create an individual that has a predicted 0% likelihood of donating. The path of many of these characteristics demonstrate an alumnus that does not have a donation history and metrics from the 3rd party institution are low as well. We can see this individual has never donated and graduated over two decades ago.

Breakdown Plot – High Likelihood of Donating Prediction



The characteristics displayed in the breakdown plot combine to create an individual that has a predicted 100% likelihood of donating. The path of many of these characteristics demonstrate an alumnus that is a very active donor, has a donation history and metrics from the 3rd party institution are high as well. We can see this individual has been an active donor for over nine years and has donated \$5,000-\$10,000 in 2023 alone.

Discussion on Final Model Chosen

The final model that was chosen was the **random forest model**. This model is able to make predictions from large sets of data by having each tree populate their own prediction. Each tree votes on their prediction, and the prediction with the highest votes is chosen. Due to this, the accuracy of this model is 93.2%.